

# Hypothesis testing revisited

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# The first test in history

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## London, 1629 – 1710 birth counts

| Year | Boys | Girls | More boys? |
|------|------|-------|------------|
| 1629 | 5218 | 4683  | 1          |
| 1630 | 4858 | 4457  | 1          |
| 1631 | 4958 | 4857  | 1          |
| ⋮    | ⋮    | ⋮     | ⋮          |
| 1710 | 4703 | 4605  | 1          |

Out of 82 years, how many years do you think the boys won ?

82 !

What are the odds this is purely due to chance ?  $\mathbb{P}(82 \text{ boy years out of } 82) = \left(\frac{1}{2}\right)^{82} \approx 2.1 \times 10^{-25}$



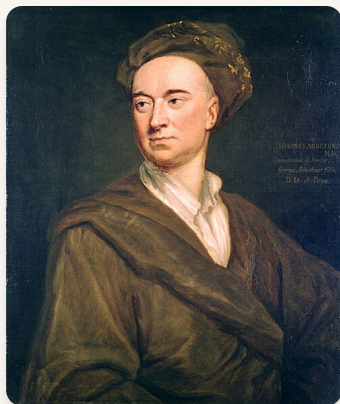
This is so unlikely. Something *must* be wrong, what could it be ?

## Is this surprising? Arbuthnot's inversion (1710)



This is so unlikely. Something *must* be wrong, what could it be ?

$$\mathbb{P}(82 \text{ boy years out of } 82) = \mathbb{P}(\text{boys}_i \geq \text{girls}_i)^{82} = \left(\frac{1}{2}\right)^{82} \approx 2.1 \times 10^{-25}$$



John Arbuthnot  
Scottish mathematician

"The assumption  $\mathbb{P}(\text{boys}_i \geq \text{girls}_i) = \frac{1}{2}$  must be wrong"

The first *probabilistic "proof" by contradiction*

# Who's the better gamer ?

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# A semester of FIFA



**Karim:** "I won **15 of our last 20** games. I'm just better."

**Yassine:** "No. You just got **lucky**."

Is 15 out of 20 *extreme* enough to *prove* that Karim is better ?

# Model the dispute

- $X_i = 1$  if Karim wins game  $i$ : Bernoulli( $\theta$ ), 20 independent games
- $\theta$  = probability that Karim wins a game; score  $S = \sum_{i=1}^{20} X_i$
- The dispute becomes a sentence about  $\theta$ :

$$H_0 : \theta = \frac{1}{2} \text{ (equals + luck)} \quad \text{vs} \quad H_1 : \theta > \frac{1}{2} \text{ (Karim is better)}$$

If *they were equals*, what is the probability of something at least this extreme?

↓ formalizing ↓

If  $H_0$  is true, what is  $\mathbb{P}(S \geq 15)$  ?

Under  $H_0$ :  $S \sim \text{Bin}(20, \frac{1}{2})$        $\mathbb{P}(S \geq 15) = \sum_{k=15}^{20} \binom{20}{k} \left(\frac{1}{2}\right)^{20} = 0.021 \leftarrow \text{the } p\text{-value}$

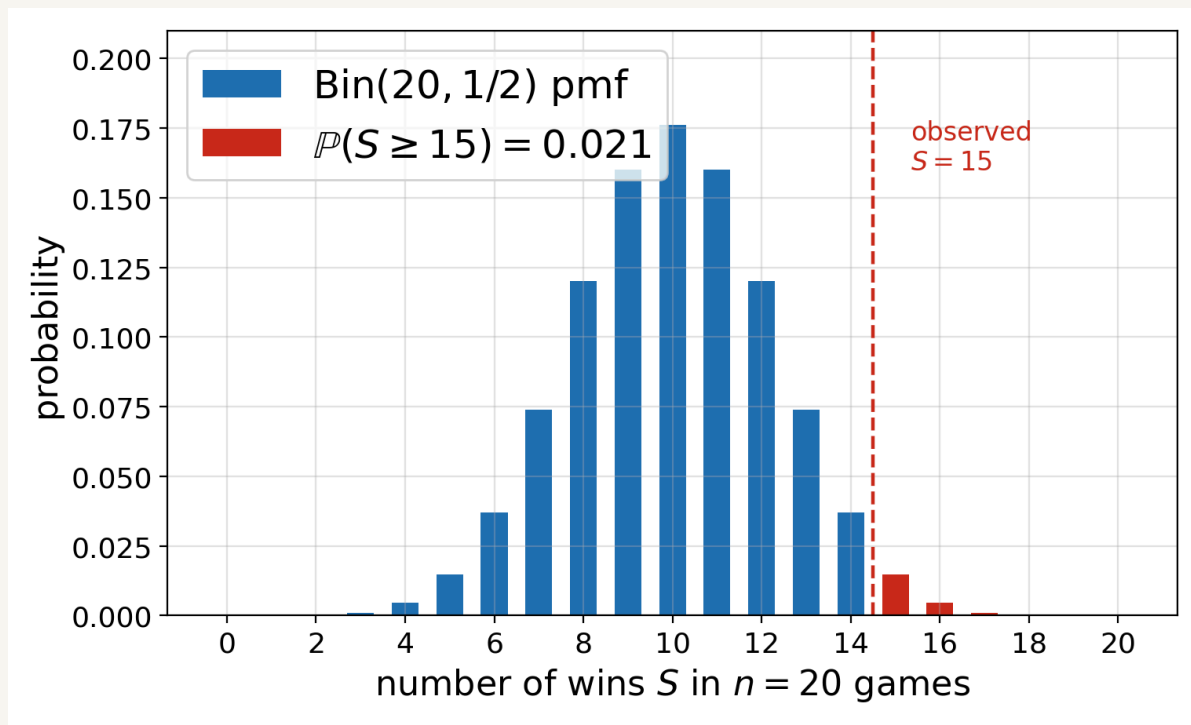


it's 2%, can't be luck

2% is a lot, not significant



# The p-value



Is 0.021 small enough to *reject*  $H_0$  ?

## The recipe, formally

0. Agree on a convention first on what is **extreme / rare enough** ?  $\alpha = 1\%$  ?  $= 5\%$  ?
1. **Hypotheses**:  $H_0$  to be challenged,  $H_1$  the hypothesis replacing  $H_0$ : the alternative.
2. **A statistic**  $S$  that gets rare when data drift from  $H_0$  toward  $H_1$ ; observed value is  $S_{\text{obs}}$
3. **Its distribution** under  $H_0$ : to be able to compute the p-value:  $p \stackrel{\text{def}}{=} \mathbb{P}(S \text{ as extreme as } S_{\text{obs}})$

Assume  $H_0 \Rightarrow$  compute  $p$

✓  $p \leq \alpha$

“Contradiction”  $\Rightarrow$  **reject**  $H_0$

✗  $p \geq \alpha$

No “Contradiction”  $\Rightarrow$  **fail to reject**  $H_0$

- Example: assume  $S = 12$ . Then, if  $H_0$  is true,  $\mathbb{P}(S \geq 12) = 0.252$ . What can we say ?
- The value  $S = 12$  is *likely* (not surprising) under  $H_0$ : does that mean  $H_0$  is true ?



**No !** We can never “accept”  $H_0$ , we either **reject** it or **fail to reject** it.

# A test can be misleading

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## Large sample size: significant but not important

- 2000 games this time; Karim wins 1050, a 52% win rate

$$p = \mathbb{P}(S \geq 1050 \mid \text{Bin}(2000, \frac{1}{2})) = 0.0134$$

- Decisive rejection. And yet  $\hat{\theta} = 0.525$ , confidence interval  $[0.503, 0.547]$
- The 52.5% difference is real, but Karim is **barely better**

With large  $n$ , tests become very sensitive to small deviations from  $H_0$ : *Powerful* test

The difference is statistically **significant** (effect is real) but **not important** (effect is small)

## Small sample size: not significant but could be important

- The opposite season: only 8 games; Karim wins 6, the **same** 75% win rate as 15 of 20

$$p = \mathbb{P}(S \geq 6 \mid \text{Bin}(8, \frac{1}{2})) = 0.145$$

- Large  $\hat{\theta} = 0.75$ , but this time the test says **not significant**
- $\hat{\theta} = 0.75$ , but the 95% confidence interval is  $[0.41, 0.93]$ : it **contains**  $\frac{1}{2}$

With small  $n$ , tests are **blind** to real deviations: the test has **low power**

The difference is **not significant** but it could be important

# How to use a statistical test ?



Two use cases:

1. Test as **strong** evidence for  $H_1$

If you reject  $H_0$  in favor of  $H_1$

with for e.g p-value  $\approx 10^{-5}$

2. Test as **red-flag** detector (weak evidence)

You want to “assume”  $H_0$

Make sure we don't reject  $H_0$

A statistical test is not a perfect decision tool:

- Large  $n$ : **negligible** deviations can get  $H_0$  rejected  $\Rightarrow$  **rejecting does not imply practical importance**
- Small  $n$ : **large** deviations go undetected  $\Rightarrow$  **failing to reject proves little**

## Exercise 1

Does the null have to be an equality ?

We now test  $H_0 : \theta \leq \frac{1}{2}$  vs.  $H_1 : \theta > \frac{1}{2}$ .

For each  $\theta \leq \frac{1}{2}$ , define  $q(\theta) = \mathbb{P}_\theta(S \geq S_{\text{obs}})$ .

1. To reject  $H_0$ , For which  $\theta$  must  $q(\theta)$  be small ?
2. What quantity should therefore define the p-value?

## Exercise 2

What counts as extreme ?

Same 15 wins out of 20, but the question changes: is Karim's win rate **different** from  $\frac{1}{2}$  ?

$$H_0 : \theta = \frac{1}{2} \longrightarrow \theta = \frac{1}{2} \quad (\text{unchanged})$$

$$H_1 : \cancel{\theta > \frac{1}{2}} \longrightarrow \theta \neq \frac{1}{2}$$

1. Which values of  $S$  should now count as extreme, and why ?
2. The pvalue was 0.021 with  $H_1 : \theta > \frac{1}{2}$ , what is it now with  $H_1 : \theta \neq \frac{1}{2}$ ?

### Exercise 3

Bonferroni correction

1. We set the significance level  $\alpha$ , and assume  $H_0$  is in fact **true**. What is the probability that a test rejects  $H_0$  ?
2. Consider  $m$  tests denoted by  $T_1, \dots, T_m$  all with true  $H_0$  each with significance level  $\alpha_i$ . Show that  $\mathbb{P}(\text{at least one } T_i \text{ rejects}) \leq \sum_{i=1}^m \alpha_i$
3. When doing  $m$  tests, what significance threshold  $\alpha_i$  can we set to guarantee that this probability remains below  $\alpha$  ?

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